DTMES 2026 · AFRICA ELECTRONICS INNOVATION CHALLENGE · BUILT STAGE

# Hydra Pay

## Fast payments. Safer transactions. Clearer records.

A low-cost ESP32 + RFID payment and transaction-recording system for small businesses that serve many customers fast: matatus, canteens and market traders.

ESP32 + RC522 | M-Pesa Daraja | WhatsApp + USSD




**Team HYDRA-PAY**

Technical University of Kenya

THE PROBLEM

# Busy small businesses lose time and money at the till

### Cash slows everything

Counting money and finding change at every sale builds queues. Cash leaves no record and is a target for theft.

### Mobile money disputes

Fake confirmation SMSs, screenshots, reversals and "I've sent it" arguments. Checking each payment holds up the queue.

### Records don't add up

Sales sit in a notebook, the till and a phone inbox. At the end of the day owners can't reconcile what they earned.

**Owners need an affordable way to accept, verify and record payments without slowing down.**

MARKET EVIDENCE

# Cash and M-Pesa live side by side

**99%**

of surveyed small businesses run partly or wholly in cash

**86%**

accepted digital payments by June 2023

**Cash dominant**

Mobile money is everywhere, but cash is still how many businesses get paid

Sources: FinAccess MSE Tracker Survey 2023; World Bank analysis of Kenya's mobile money market.

**Hydra Pay works with the M-Pesa habits people already have. The opportunity is making everyday transactions easier to manage.**

OUR SOLUTION

# One tap: pay, verify and record in one flow

1

## Tap

The customer taps a card, key tag or wristband on the reader at the counter or matatu door.

2

## M-Pesa prompt

The price is sent as a Daraja STK push to the customer's phone, paid to the business till.

3

## Verify

The customer enters their PIN. Hydra Pay marks it PAID only when Safaricom confirms.

4

## Record

Saved with amount, time and M-Pesa code. WhatsApp receipt sent, dashboard updated live.

**Pay-per-tap:** every tap sends one M-Pesa prompt for the price.

**Prepaid wallet:** regulars top up once, each tap deducts instantly.

PROJECT OBJECTIVES

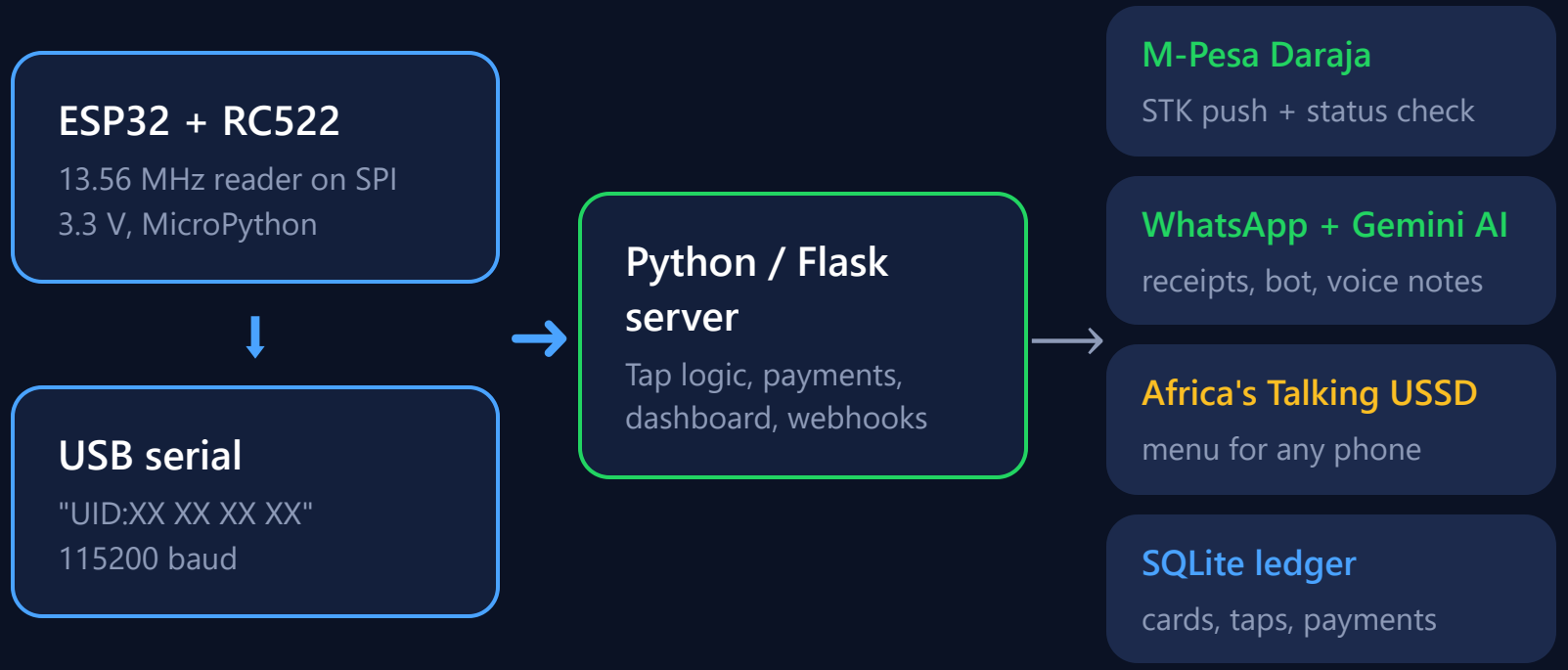
# What we set out to build, and where we are

**Main objective:** an affordable, easy-to-use payment and transaction-recording system that improves payment speed, traceability and recordkeeping for high-volume small businesses.

- ✓ Design a streamlined payment workflow for fast-paced businesses (built)
- ✓ Develop a reliable way to confirm completed payments (built)
- ✓ Record and retrieve transaction details automatically (built)
- ✓ Give simple daily summaries for reconciliation (built)
- → Reduce the time to complete and verify a transaction (measure in pilot)
- → Evaluate affordability, usability, reliability and security with real businesses (pilot)

TECHNICAL DESIGN & SOUNDNESS

# How it is built




## Engineering decisions

**Reliable reads:** firmware debounce, wake-up of half-read cards, 3× retry and max antenna gain (48 dB).
**Verified money:** PAID only on Safaricom's answer (callback, with status query as backup), recorded exactly once under a lock.
**Auditable:** append-only ledger; undo is a reversal entry, nothing is deleted. CSV export.
**Secure admin:** PIN for cards, cash and undo; remote login blocked while the PIN is default.
**Self-healing:** finds the USB port and reconnects if the reader is unplugged.

FUNCTIONALITY / PROOF OF WORK

# It works today, and you can try it at our table

**27**
M-Pesa prompts sent through Daraja

**10**
payments confirmed PAID by Safaricom

**15**
declined or unanswered, correctly NOT PAID

Daraja sandbox test runs, 22–23 Oct 2026

## Live demo

Tap a card: watch the M-Pesa prompt arrive, the dashboard update and the WhatsApp receipt land.

## Working features

- ✓ Card and key-tag taps read and logged as IN / OUT
- ✓ STK push to the card owner's phone for the card's price
- ✓ WhatsApp receipt with the M-Pesa code after payment
- ✓ WhatsApp bot: my card, my payments, number change
- ✓ AI assistant in English, Kiswahili and Sheng, plus voice notes
- ✓ Live dashboard: register cards, daily totals, CSV export
- ✓ Lost card: move the account to a new card in one click
- ✓ USSD menu on any phone via Africa's Talking

TARGET MARKET

# Businesses where every second counts

PILOT FIRST

## Canteens & cafeterias

Lunch-hour rush, fixed prices, regular customers.

**Need:** shorter checkout and accurate sales records.

PILOT FIRST

## Market & retail traders

Mixed cash and M-Pesa every day.

**Need:** easy payment tracking without expensive software.

NEXT

## Matatus & transport

Many small fares under time pressure. Tap IN to board.

**Need:** fast fare collection, confirmed fares, clear daily takings.

**Customer:** the business owner or operator. **Users:** the person paying and the person receiving. We start where we can observe and measure, then move to matatus.

UNIQUE VALUE PROPOSITION

# Less friction at the till, clearer records after it, on one low-cost system

| TODAY | WHAT IT ALREADY DOES | WHAT HYDRA PAY ADDS |
|---|---|---|
| **Cash** | Familiar and immediate | No change to count, every sale recorded automatically |
| **M-Pesa till / Paybill** | Widely used digital payments | No typing a till number or amount: tap, then PIN. Fake SMSs don't count |
| **M-Pesa Business Hub** | Histories, reports, reconciliation | Tap-to-pay hardware plus a lightweight flow built for queues and fares |
| **Sales notebook** | Cheap recordkeeping | No manual entry; searchable records, daily totals and CSV |

**Speed** for peak hours

**Confidence:** Safaricom-verified

**Automatic** records

**Affordable** hardware

BUSINESS MODEL

# Affordable for the seller, sustainable for us

### Freemium

Basic transaction recording free; advanced reports and features paid.

### Small subscription

An affordable weekly or monthly fee per reader.

### Business packages

Fleet and multi-stall dashboards for SACCOs and larger vendors.

### Partnerships

Matatu SACCOs, market managers, canteen operators and business associations.

**No cut of each sale: low-value, high-volume sellers keep their margin. We'll test willingness to pay in the pilot before fixing prices.**

FEASIBILITY, SECURITY & RISKS

# Risks we've designed for

| CHALLENGE | HOW HYDRA PAY HANDLES IT |
|---|---|
| **Fake confirmations** | Status comes only from Safaricom's Daraja API. An unverified payment is never shown as PAID |
| **Reversals & disputes** | Every payment keeps its M-Pesa code, status and time in an append-only audit trail; disputes follow Safaricom's process |
| **Slow checkout** | Tap, then PIN. No typing. Checkout time will be measured in the pilot |
| **Poor network** | Pending is kept separate from paid, and "still processing" is never marked failed. Offline tap queue is next |
| **Tight budgets** | Off-the-shelf ESP32 + RC522, no POS terminal, no per-sale fee |
| **Sensitive data** | Admin PIN, keys kept out of code, minimal data stored; aligned with Kenya's Data Protection Act |

ROADMAP & VALIDATION

# From working prototype to real businesses

DONE

## Prototype

Reader, M-Pesa, verification, records, WhatsApp, USSD on the Daraja sandbox.

NEXT

## Discovery

10–20 interviews with vendors and conductors about payments, disputes and records.

NEXT

## Go live

Live Safaricom till, live USSD short code, offline reader that syncs later.

PILOT

## Controlled pilot

A few canteens and traders, measured against a before baseline.

LATER

## Expand

Matatu SACCOs, then multi-currency. The owner always gets KES.

**We'll measure:** transaction time · queue time at peak · % of transactions recorded correctly · verification success rate · reconciliation errors · merchant satisfaction and willingness to pay. Targets are set after the baseline.

INDUSTRY ENGAGEMENT

# Meeting Safaricom leadership



With Safaricom's former Chief Financial Services Officer



With Safaricom's Chief Technology & Information Officer

# Tap. Pay. Done.

Hydra Pay: fast payments, safer transactions, clearer records.

**Questions? Try the live demo**

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